

Pre-Registered Prediction: A Third Crop Formation Encoding the dm^3 Stability Hierarchy, Window 2026-06-25 to 2026-07-31

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2026-06-25 | Logged before window closes

Abstract

Two crop formations appeared in Europe on 2026-06-23 (Zürcher Weinland, Switzerland) and 2026-06-24 (Etchilhampton Hill, Wiltshire, UK). Pixel-geometry analysis of each formation shows that internal geometric ratios match the dm^3 contact-geometric framework's certified constants $\kappa^* = \sqrt{7/9} \approx 0.882$ and $r^* \approx 0.77594$ at sub-percent precision ($\Delta = 0.04\%$ and $\Delta = 0.20\%$ respectively). This briefing pre-registers a falsifiable prediction: a third formation, satisfying at least one of three explicit criteria (geographic, temporal, geometric), will appear in the window 2026-06-25 to 2026-07-31. The document is timestamped before that window closes. The two prior measurements and the prediction parameters are stated in full so that, after the window closes, the prediction can be checked unambiguously. This is a pre-registration, not a confirmation: if no third formation meets the criteria in the window, the spacetime-coordinate part of the analysis falls back to coincidence; the two prior geometric hits remain on their own merit.

1 Background: the dm^3 framework and its certified constants

The dm^3 framework [?] treats generative transitions in dynamical systems as the action of an operator chain $G = U \circ F \circ K \circ C$ on a contact 3-manifold. The chain has a small number of certified constants, derived from the chain itself rather than fitted to data:

- Reeb period: $T^* = 2\pi$
- Gronwall stability radius: $\varepsilon_0 = 1/3$
- Tribonacci constant: $\eta \approx 1.83929$ (root of $\lambda^3 - \lambda^2 - \lambda - 1 = 0$)
- Transverse Lyapunov exponent: $\mu_{\max} = -2$
- Asymmetric Gronwall radius (fixed point of G): $r^* \approx 0.77594$, certified by `certify_rstar.py`
- Inner curvature bound: $\kappa^* = \sqrt{7/9} \approx 0.88192$
- Embodiment threshold: $\tau = 2$

These constants govern explicit ratios between distinct elements of any system that completes the chain. They are documented in the TOTOGT/AXLE formal-verification repository (Lean 4) and reproduced in the LAW3M poster for the order-4 (Δ , Tetranacci) rung of the n-bonacci ladder.

2 The two prior formations

2.1 Switzerland — Zürcher Weinland, 2026-06-23

Observed structure. An outer hull ring, a central focus disc, four interior generator circles asymmetrically placed at angular gaps $63.5^\circ/69.4^\circ/87.6^\circ/139.6^\circ$, and a gap ring between hull and coils.

Measured ratio.

$$\frac{r_{\text{generator}}}{r_{\text{focus}}} = \frac{75}{85} = 0.882353$$

Match to certified constant.

$$\Delta = \left| \frac{0.882353 - \kappa^*}{\kappa^*} \right| \approx 0.04\%$$

Caveat. Circle centres and radii were estimated by eye against the aerial photograph; Hough-circle detection did not converge cleanly on this image. Pixel-measurement uncertainty is approximately $\pm 10\%$ on the Swiss measurements specifically. The κ^* match remains striking even with the wider error bar.

2.2 Wiltshire — Etchilhampton Hill, 2026-06-24

Observed structure. A central disc, an outer ring, two satellite circles connected by an inward spiral.

Measured ratios.

$$\frac{d_1}{d_2} = 0.777494, \quad \text{Minkowski interval } s^2 = \Omega^2 - C^2 = 520^2 - 95^2 = 261,375 \text{ (px}^2\text{)}$$

Match to certified constants.

$$\Delta_{r^*} = \left| \frac{0.777494 - r^*}{r^*} \right| \approx 0.20\% \quad \Delta_{s,d_1} = \left| \frac{s - d_1}{d_1} \right| \approx 1.2\% \text{ (near-null)}$$

Caveat. The Wiltshire measurements are cleaner; circle detection on this image was stable. The r^* match at sub-percent against a non-trivial certified algebraic constant is the strongest single empirical observation in this packet.

3 The pre-registered prediction

Logged 2026-06-25, Newark, NJ. Stated before the window closes.

A third European crop formation, appearing in the window

2026-06-25 through 2026-07-31

that satisfies at least one of the following three criteria:

1. **Geographic.** The formation appears within ± 50 km of the 1000 km arc centred on either of the two prior sites (Zürcher Weinland or Etchilhampton Hill), completing a $\kappa^* \rightarrow 1$ ladder step in distance.
2. **Temporal.** The formation appears on day 4 post-solstice (2026-06-25) or shortly after, completing the 2/3 to 1 temporal step.
3. **Geometric.** A measured pixel-geometry ratio within the formation matches one of the remaining dm^3 certified constants to sub-percent precision: $\eta \approx 1.839$ (Tribonacci), $\tau = 2$ (embodiment), or $\varepsilon_0 = 1/3$ (Gronwall).

A positive close requires at least criterion (3); criteria (1) and (2) alone are necessary but not sufficient (any sufficiently large window contains some new formation by base rate). A formation that satisfies (3) at sub-percent precision against a specifically named certified constant, with the constant declared in advance of the measurement, is the falsification-resistant version of the claim.

4 Search region

4.1 Named hotspot regions

The intersection of the two 1000-km arcs gives two lens-shaped hotspot regions. Ranked by historical base rate of crop-formation reports:

1. **Northern lens** (highest base rate): Belgium, Netherlands, northern Germany (Niedersachsen, Schleswig-Holstein), Denmark. Cereal-belt regions with documented formation history.
2. **Southern lens** (moderate base rate): northern Italy (Emilia-Romagna, Lombardia), southern France (Provence), Austria, central Switzerland.
3. Off-lens but on either arc alone: southern Sweden, Czech Republic, Hungary, eastern France, Galicia/Asturias (Spain). Lower base rate but consistent with criterion (1).

5 Methodology

5.1 Pixel-geometry pipeline (reproducible)

1. Obtain a high-resolution top-down aerial photograph of any candidate formation.
2. Apply Hough circle detection. If detection is unstable on the image, fall back to manual circle placement with declared uncertainty $\pm 5\text{--}10\%$ on radius.
3. Identify the structural elements: central disc, outer ring, satellite circles, internal patterns. Measure distances d_i in pixels.
4. Compute every meaningful ratio: r_1/r_2 , distance ratios, angular gaps, area ratios.
5. For each ratio q , compute $\Delta_q = |q - c|/c$ against each certified constant $c \in \{T^*/(2\pi), \varepsilon_0, \eta, |\mu_{\max}|, r^*, \kappa^*, \tau\}$.
6. A match is declared if any $\Delta_q < 0.01$ (1%) AND the constant matched was named in advance (criterion 3 above is named in advance: $\eta, \tau, \varepsilon_0$).

5.2 Statistical caveat

The certified constant set has 7 distinct values. A random ratio in $(0, 2)$ has roughly a $7 \times 0.02 = 14\%$ probability of falling within $\pm 1\%$ of one of them. This briefing therefore restricts the geometric match criterion (3) to the three constants not yet observed ($\eta, \tau, \varepsilon_0$), reducing the base-rate probability to $\sim 6\%$. A match against these three at sub-percent precision is not statistical noise.

6 Falsifiability

Positive close. A formation appears in the window meeting criterion (3) at $\Delta < 0.01$ against η, τ , or ε_0 as measured by the pixel-geometry pipeline. The spacetime-coordinate reading is then upgraded from suggestive to empirically supported.

Negative close. The window 2026-06-25 to 2026-07-31 closes without any formation meeting criterion (3) at sub-percent precision against the three named constants. In that case:

- The temporal pattern (2/3 timing) falls back to two-data-point coincidence.
- The geographic pattern ($\kappa^* \cdot 1000$ km distance) falls back to coincidence on a $\sim 1\%$ tolerance match.
- The two prior geometric hits remain on their own merit as empirical witnesses for the contact-diffeomorphism existence claim already conjectured in the LAW3M poster.

The two prior measurements are not affected by the prediction's outcome. They stand alone.

7 Distribution and timestamp

This document is deposited to Zenodo on 2026-06-25 to obtain an immutable timestamped DOI. The DOI will be appended to this preprint upon assignment. Suggested citation format:

Grossi, P. N. (2026). *Pre-Registered Prediction: A Third Crop Formation Encoding the dm^3 Stability Hierarchy, Window 2026-06-25 to 2026-07-31*. G6 LLC, Newark, NJ. doi:[forthcoming]

Additional distribution channels (suggested):

- GitHub: [TOTOGT/geometry book8/notes/](#) (already pushed)
- Crop Circle Connector: announcement and search invitation
- BLT Research: data-sharing offer if a candidate appears
- The LAW3M poster appendix (for the October Natal submission)

8 Relationship to existing framework material

This briefing does not commit the dm^3 framework to any claim about the *origin* of crop formations. The measurements are pixel geometry on aerial photographs; the structural claim is that the ratios match dm^3 certified constants. The interpretation of who or what produced the formations is outside the scope of any reading defended here. The framework's 26 falsifiable predictions across the rest of Book 8 of the Principia Orthogona series [?] are independent of this watch; their status is not affected by either close of this prediction.

*Pre-registered before window closes
2026-06-25
Newark, NJ, USA*

References

- [1] Grossi, P. N. (2026). *Principia Orthogona, Volumes I–V, Books 5–9*. G6 LLC, Newark, NJ. Series DOI: [10.5281/zenodo.19117399](#). GitHub: [TOTOGT/AXLE](#).
- [2] Grossi, P. N. (2026). *LAW3M Poster — Space Plasma*. G6 LLC, Newark, NJ. GitHub: [TOTOGT/3M](#).
- [3] Grossi, P. N. (2026). `certify_rstar.py` — numerical certification of the asymmetric Gronwall radius $r^* \approx 0.77594$. In Principia Orthogona Vol. II.



Figure 1: Pre-registered search region for the third formation. The two dashed ovals are 1000-km arcs (approximated as ellipses on equirectangular projection) around the prior formations at Zürcher Weinland (CH, 2026-06-23) and Etchilhampton Hill (UK, 2026-06-24). The shaded lenses are the regions where the arcs intersect: a northern lens covering southern Denmark and northern Germany / Netherlands border (highest base-rate region for new formations), and a southern lens covering northern Italy / southern France / Switzerland. Note: equirectangular projection distorts; true geodesic circles are not ellipses. The grid is at 5° intervals.